INNOPOLIS UNIVERSITY

SNE MASTER TRACK

2025

COURSE: [F25] Networks Engineering

INR LAB 1 Report

On

Networking Basics

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Submission Date: 24.11.2025

Task 1 - Tools

To make it easier to prototype network topologies and troubleshoot them, you will be using GNS3 software to emulate networks.

1. Install the needed dependencies for GNS3: QEMU/KVM, Docker and Wireshark. Hint: Check that virtualization is enabled in your bios. Make sure that the user belongs to all the necessary groups after installing the tools.

2. Start a GNS3 project, configure the pre-installed Ubuntu Cloud Guest template. Check that you can start it.

3. What are the different ways you can configure internet access in GNS3? Test them with a single created VM and give a one-line description of each.

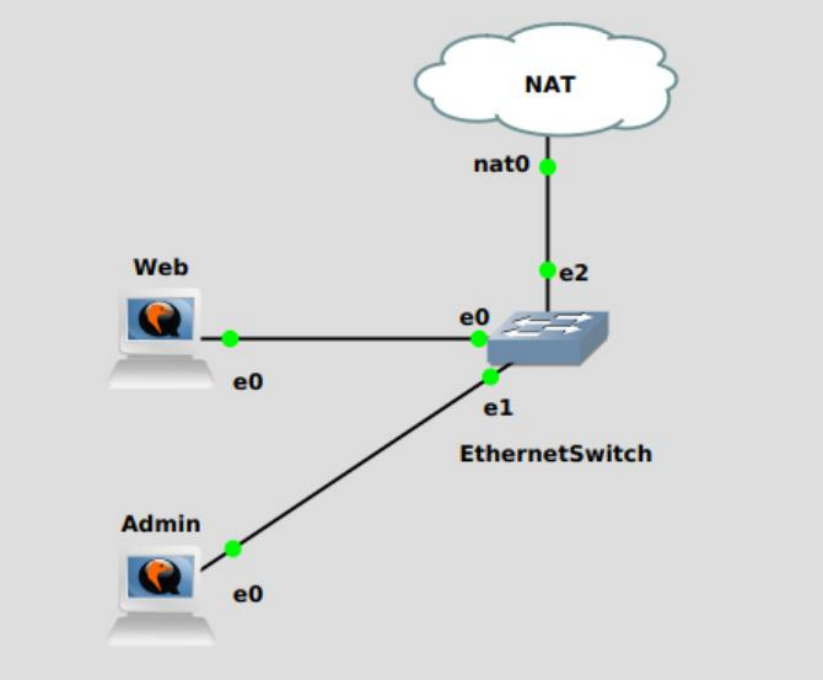
What are the differences between them?

Bonus: show the difference between them on practice and test the connectivity.

Task 2 - Switching

Now that you have the necessary tools and are familiar with GNS3, you will deploy your first network.

1. Make the following network topology:



2. Install openssh-server on both VMs and nginx web server on the Web VM.

3. What is the IP of the mask corresponding to /28 ?

How many machines can you configure under this subnet? Explain it.

4. Configure the VMs with private static IPs under a /28 subnet.

5. Check that you have connectivity between them.

Hint: use ping, traceroute and mtr.

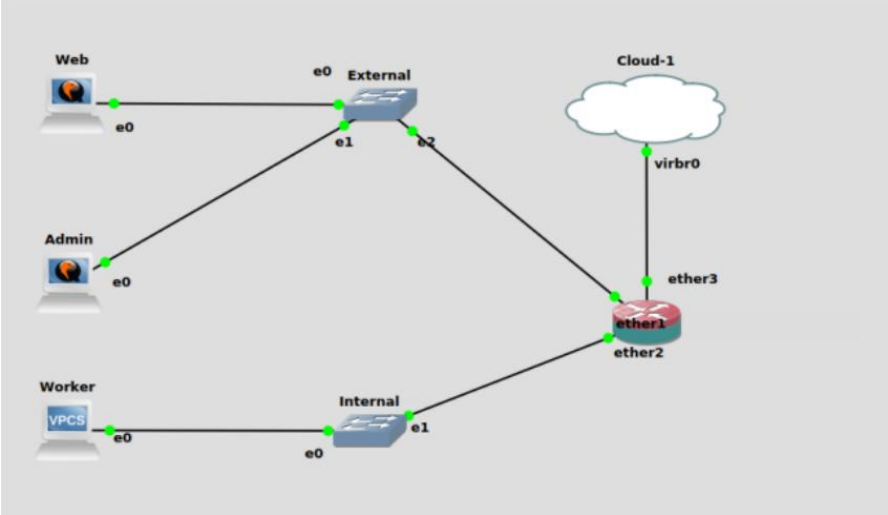
6. Make sure your web server is accessible from the Admin VM.

Hint: use curl or wget

Task 3 - Routing

Now that you have a small network running, it is time to have it properly routed. Delete the NAT device because you will make your own gateway now.

1. Select a virtual Routing solution (Gateway) such as Mikrotik (recommended default choice), PfSense, VyOS, Untangle NG, OpenWrt, Cumulus VX.



2. Create Internal network for Worker instance.

3. Connect your Gateway to the internet and to your workstation/host.

4. Setup the gateway for Admin, Web and Worker, then check their connectivity.

5. Configure port forwarding for http and ssh to Web and Admin respectively.

6. Check that you can ssh to the Admin and access your web page from your workstation/host.

Solutions:

Task 1 – Tools

1.1: Installing the needed dependencies for GNS3:

Check if Virtualization is enabled (BIOS)

To run VMs quickly in GNS3, KVM is required. So, it needs to know whether the CPU's virtualization is enabled or not. GNS3 uses KVM to run virtual machines quickly, so virtualization must be enabled on the CPU [1]. To verify this in Ubuntu, run the following command in the terminal, where Virtualization enabled is found [2].

```
hasib@hasib-ThinkPad-T480s:~$ lscpu | grep Virtualization
Virtualization:                       VT-x
hasib@hasib-ThinkPad-T480s:~$
```

Go to BIOS and under Security, there is an option called Intel VT-x and enable it.

Installing the required packages and updating the system:

```
hasib@hasib-ThinkPad-T480s:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for hasib:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 https://dl.google.com/linux/chrome/deb stable InRelease [1,825 B]
Get:4 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 https://dl.google.com/linux/chrome/deb stable/main amd64 Packages [1,206 B
]
Get:7 https://ppa.launchpadcontent.net/gns3/ppa/ubuntu noble InRelease [18.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [37.1
 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,3
17 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,619
 kB]
Get:12 https://ppa.launchpadcontent.net/gns3/ppa/ubuntu noble/main amd64 Package
s [1,848 B]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [216
 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [2
```

QEMU/KVM installation [2]:

```
hasib@hasib-ThinkPad-T480s:~$ sudo apt install qemu qemu-kvm libvirt-daemon-syst
em libvirt-clients bridge-utils virt-manager -y
```

Docker installing:

```
hasib@hasib-ThinkPad-T480s:~$ sudo apt install docker.io -y
```

Enabling Docker [4]:

```
hasib@hasib-ThinkPad-T480s:~$ sudo systemctl enable --now docker
```

Installing Wireshark [5]:

```
hasib@hasib-ThinkPad-T480s:~$ sudo apt install wireshark -y
```

Adding the user to the required group

The user has been added to the group for using KVM, libvirt, docker, wireshark [5].

```
hasib@hasib-ThinkPad-T480s:~$ sudo usermod -aG libvirt $(whoami)
sudo usermod -aG libvirt-qemu $(whoami)
sudo usermod -aG kvm $(whoami)
sudo usermod -aG wireshark $(whoami)
sudo usermod -aG docker $(whoami)
```

Then the computer was restarted.

1.**2: Starting a GNS3 project and configuring the pre-installed Ubuntu Cloud Guest template**

GNS3 is a Network Simulation/Emulation platform that allows you to create complete network topologies by adding VMs, Routers, etc. [6].

GNS3 Installation

The following command is written for GNS3 installation:

```
hasib@hasib-ThinkPad-T480s:~$ sudo apt install gns3-gui gns3-server -y
```

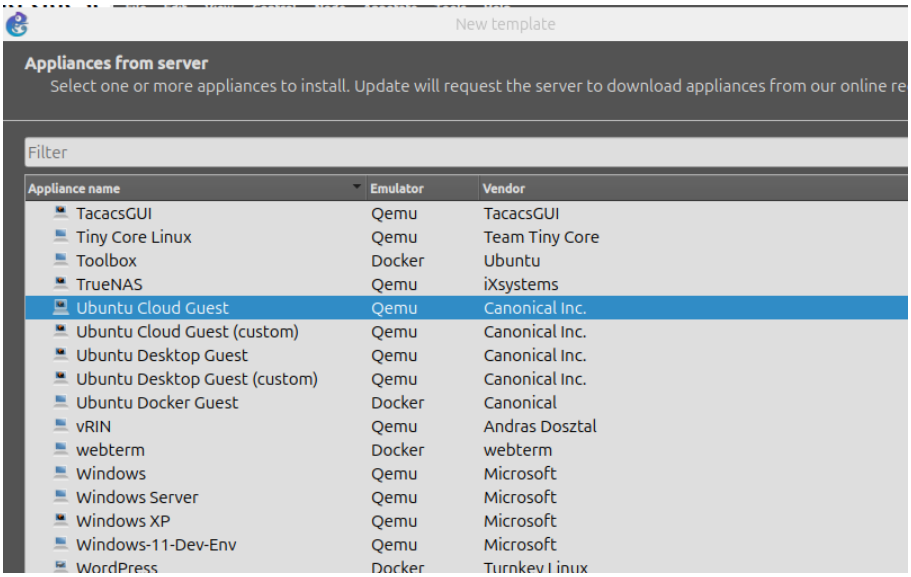
After the installation is complete:

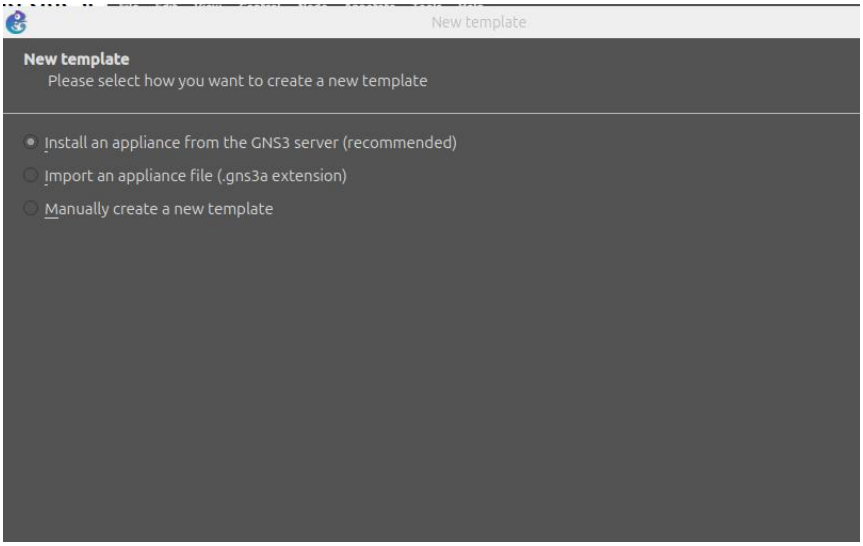
Configuring the Ubuntu Cloud Guest Template

When you run GNS3 for the first time, a window appears, from there select “Run appliances locally”, click Next and Finish to see the GNS3 workspace.

Adding Ubuntu Cloud Guest Template

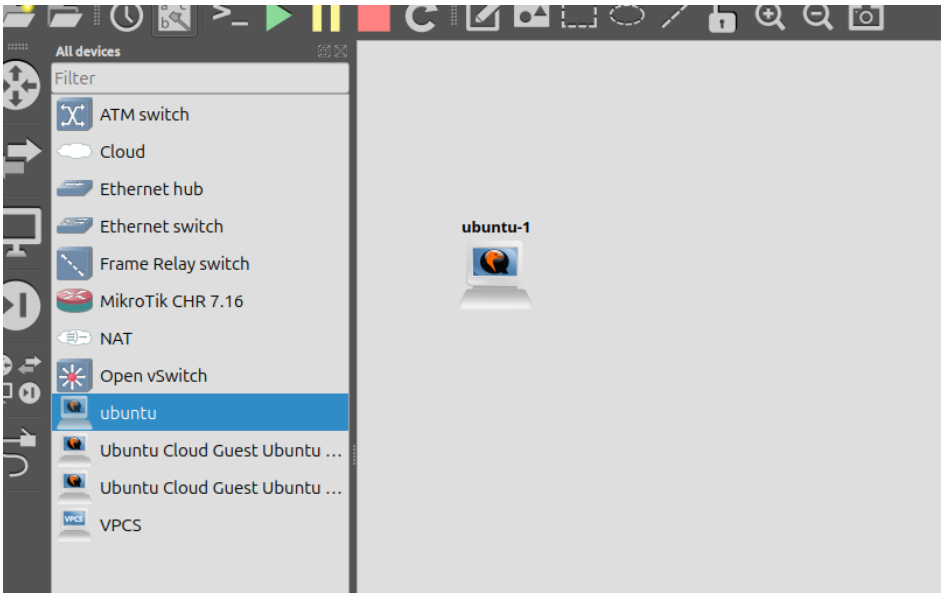
After opening the Devices panel on the right side of GNS3, I have gone to End Devices, right-click on QEMU VMs, and clicked New Template. Then I have selected Install an appliance from the GNS3 server and selected Ubuntu Cloud Guest from the list. Then I have downloaded the vmdk file, uploaded it to the location, set the RAM and CPU of the VM to 1GB RAM and 1 CPU, and then saved.

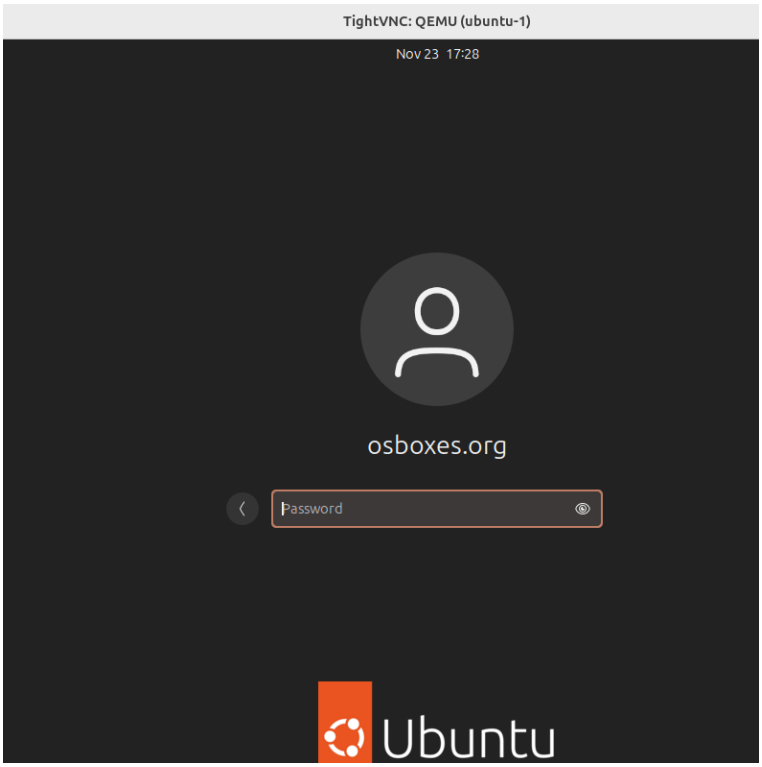




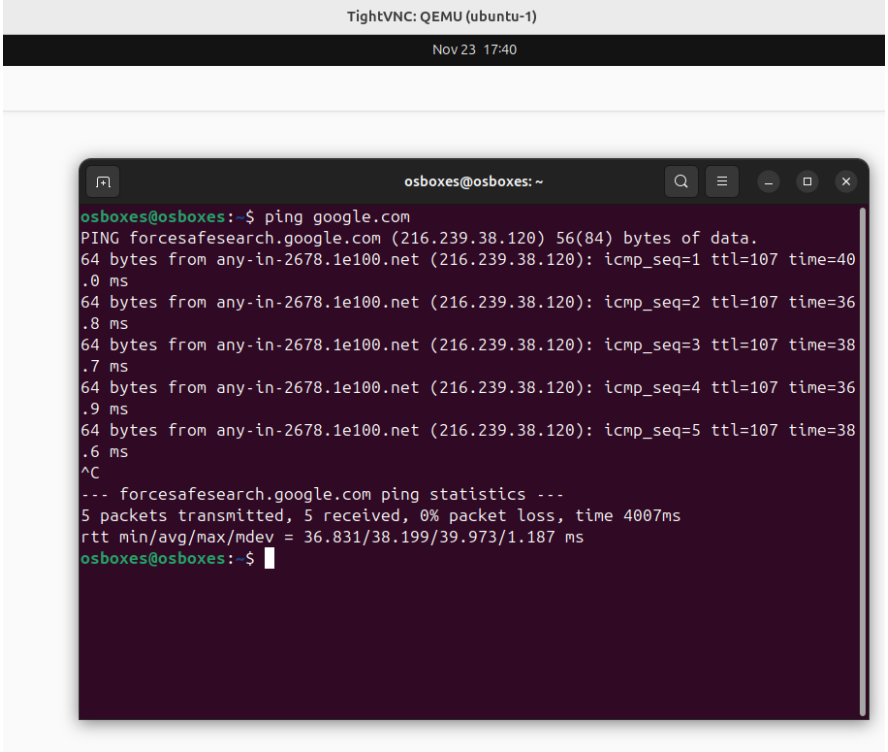
Testing the VM Startup

After dragging the Ubuntu VM icon to the workspace, clicking the Start/Play Button (Green Button) from the top boots the VM and double-clicking opens the console and shows the login screen.





I checked if internet connectivity was ok by going to Terminal and pinging google.com.

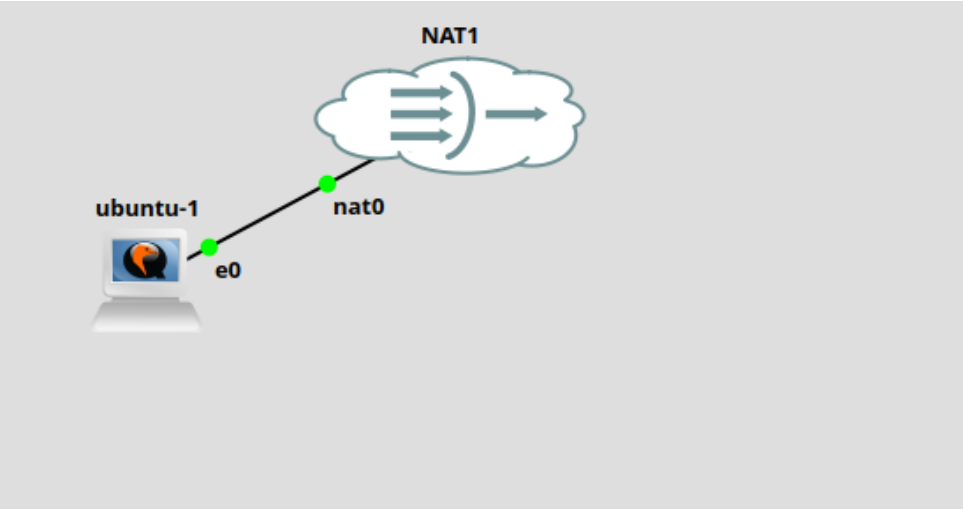


1.3: different ways of configuring internet access in GNS3

There are three common methods for providing internet to VMs in GNS3 [6]:

- **NAT (Network Address Translation)**

This is GNS3's built-in NAT node and is very easy to install. Host Internet plays a key role. The VM always gets internet, but the VM cannot be accessed from outside.



- **Cloud Node (bridged mode)**

Here, a real network card is used, where the VM is directly connected to the router/internet network. The VM can also get a real IP and it is possible to connect to the VM from the outside. As a result, Cloud Node connects the VM to your real LAN, making it visible to the real network [5], [6].

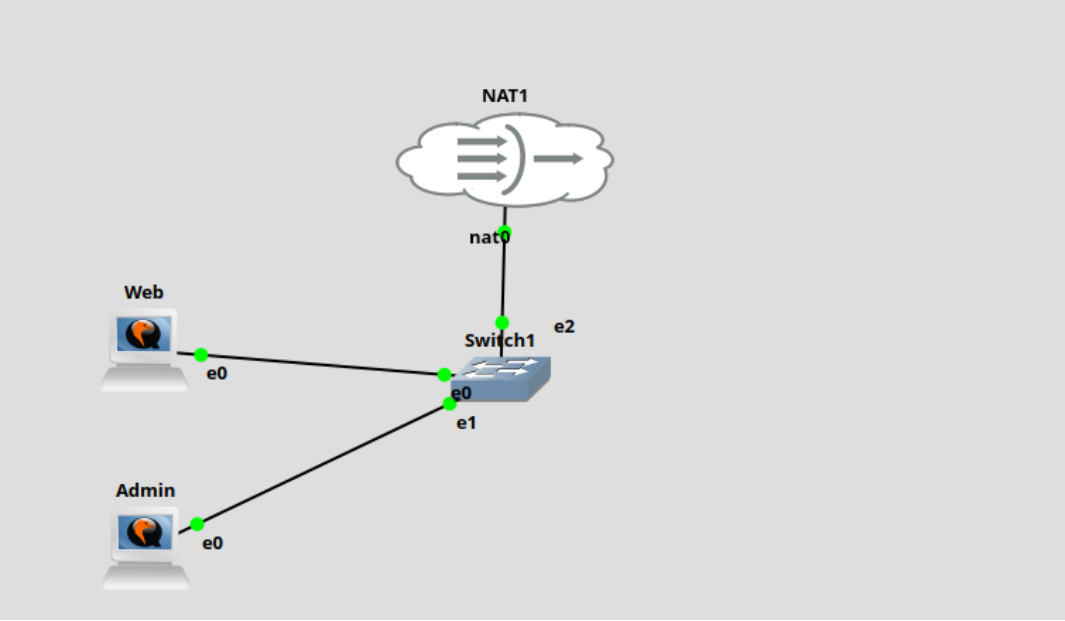
- **Host-Only**

It does not provide internet access, which is mainly used for VM to Host communication and lab testing. There is no access to the outside or the internet. Host-only networks only provide communication between the VM and the Host [5], [6].

Task 2 - Switching

2.1. Making the following network topology:

In this lab, the network topology consists of two Ubuntu VMs connected to a switch. After launching GNS3 in Ubuntu, opening the Devices panel and dragging the Ubuntu Cloud Guest template to the workspace. After Renaming the two VMs as Web and Admin, I dragged the Ethernet Switch from Devices to the middle of the workspace [7]. Then, I clicked the Add a link icon from the toolbar, clicked on the Web VM, and added it to eth0 / e0 as the interface. Similarly, I clicked on Switch and select Ethernet0/0 (e0) to connect the Admin VM to the Switch. Then I runed the topology.



2.2. Installing openssh-server on both VMs and nginx web server on the Web VM:

SSH can be used to log in to the Web VM system from the Admin VM. OpenSSH is a standard tool for secure remote login and file transfer [8]. Then, the two VMs were started.

openssh-server was installed

The following commands were run on both VMs

```
osboxes@osboxes:~$ sudo apt install openssh-server -y
```

Checked to see if the service is running:

```
osboxes@osboxes:~$ sudo systemctl start ssh
```

Then the status was checked:

```
osboxes@osboxes:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Sun 2025-11-23 17:56:02 EST; 33s ago
 Invocation: 36c8a401fc29421d876a9dad727dacfc
       Docs: man:nginx(8)
    Process: 3931 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exite
    Process: 3932 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status
   Main PID: 3963 (nginx)
      Tasks: 2 (limit: 1821)
     Memory: 2.3M (peak: 5.1M)
        CPU: 40ms
     CGroup: /system.slice/nginx.service
             ├─3963 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─3966 "nginx: worker process"

Nov 23 17:56:02 osboxes systemd[1]: Starting nginx.service - A high performance web server and a
Nov 23 17:56:02 osboxes systemd[1]: Started nginx.service - A high performance web server and a
osboxes@osboxes:~$
```

Status active (running)" is displayed [8], [9].**Installing nginx web server on Web VM**

On Web VM:

```
osboxes@osboxes:~$ sudo apt install nginx -y
Installing:
  nginx

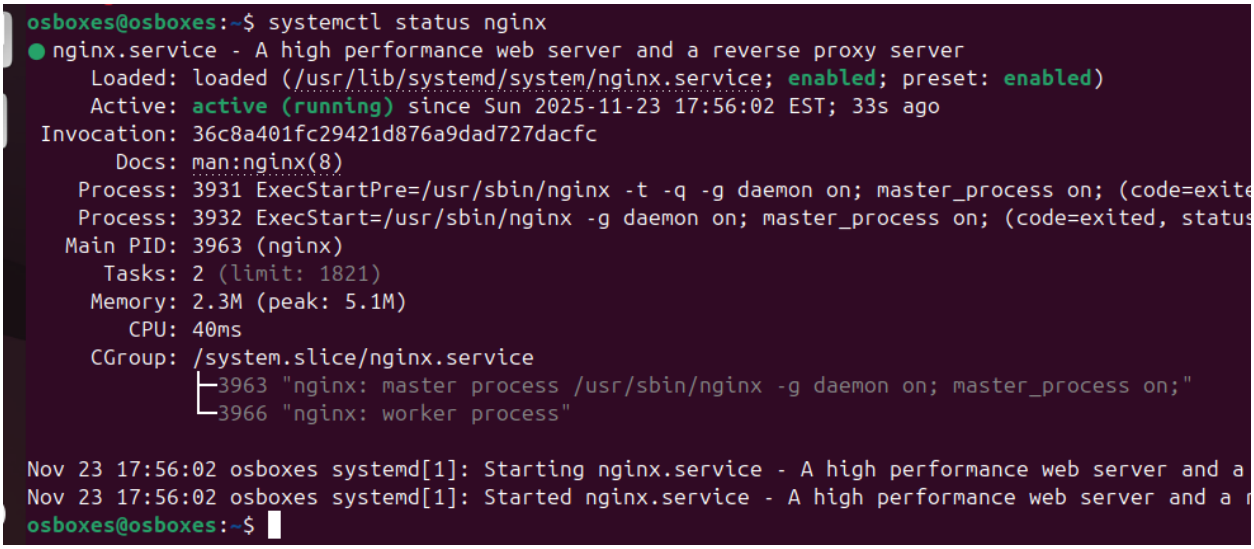
Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap  nginx-doc

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 314
  Download size: 741 kB
  Space needed: 2,108 kB / 490 GB available

Get:1 http://archive.ubuntu.com/ubuntu plucky-updates/main amd64 nginx-common all 1.26.3-2ubuntu1.
Get:2 http://archive.ubuntu.com/ubuntu plucky-updates/main amd64 nginx amd64 1.26.3-2ubuntu1.2 [69
Fetched 741 kB in 2s (364 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 143627 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.26.3-2ubuntu1.2_all.deb ...
Unpacking nginx-common (1.26.3-2ubuntu1.2) ...
Selecting previously unselected package nginx.
```

After installation, status of nginx has been checked.



Status appears active [10].

2.3 IP of the mask corresponding to /28

Subnet mask of /28

The CIDR notation /28 means that the first 28 bits of a 32-bit IPv4 address are the network portion. The remaining 4 bits are the host portion [11].

/28 subnet mask: 255.255.255.240

Total addresses and usable hosts

Total number of addresses: $2^{32-28} = 24 = 16$ addresses

Out of these-

- 1 Network address
- 1 Broadcast address
- The remaining $16 - 2 = 14$ usable hosts which are usable for devices.

/28 subnet can accommodate 14 devices, which is also confirmed in subnet calculators and subnet cheat sheets [11].

2.4. Configuring the VMs with private static IPs under a /28 subnet

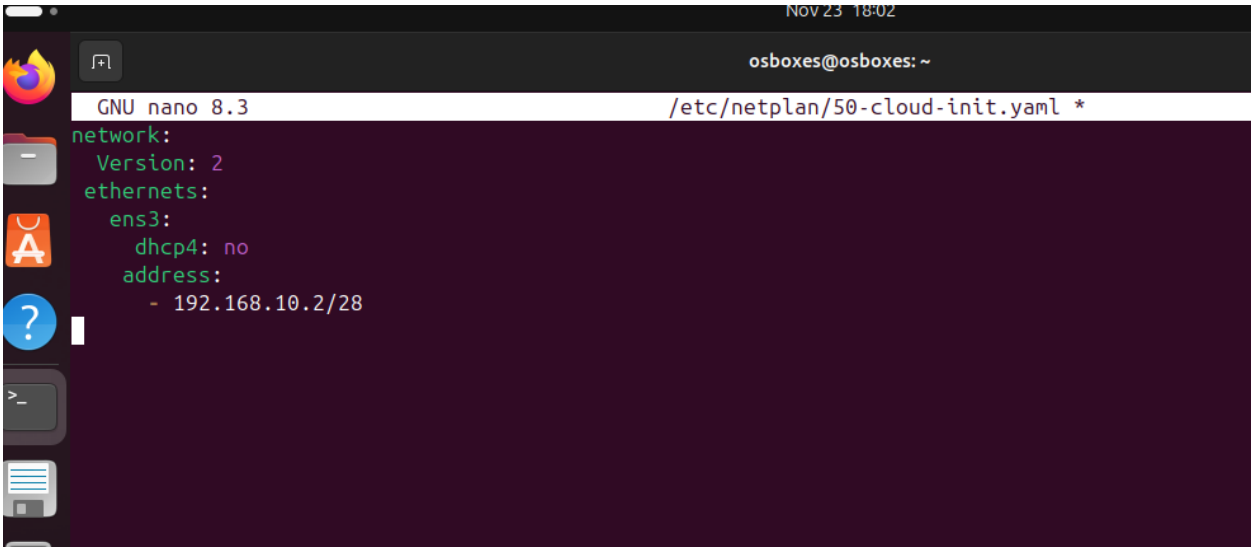
At Web VM:

The 50-cloud-init.yaml file is opened for editor to apply netplan [12], [13].

```
osboxes@osboxes:~$ sudo nano /etc/netplan/50-cloud-init.yaml
```

The file is written as:

Here, the interface is the ens3:



Apply:

After saving, the netplan has been applied.

```
osboxes@osboxes:~$ sudo netplan apply

** (generate:5472): WARNING **: 18:10:11.723: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Ne
plan configuration should NOT be accessible by others.

** (generate:5472): WARNING **: 18:10:11.725: Permissions for /etc/netplan/50-cloud-init.yaml are too open. Netplan con
iguration should NOT be accessible by others.

** (process:5470): WARNING **: 18:10:13.466: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Net
lan configuration should NOT be accessible by others.

** (process:5470): WARNING **: 18:10:13.467: Permissions for /etc/netplan/50-cloud-init.yaml are too open. Netplan conf
guration should NOT be accessible by others.

** (process:5470): WARNING **: 18:10:14.652: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Net
lan configuration should NOT be accessible by others.

** (process:5470): WARNING **: 18:10:14.655: Permissions for /etc/netplan/50-cloud-init.yaml are too open. Netplan conf
guration should NOT be accessible by others.
Show Apps boxes:~$
```

Verify:

In the Web VM, it can be seen that the 192.168.10.2/28 IP is used.

```
osboxes@osboxes:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:a5:17:8e:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    altname enx0ca5178e0000
    inet 192.168.10.2/28 brd 192.168.10.15 scope global noprefixroute ens3
       valid_lft forever preferred_lft forever
    inet6 fe80::ea5:17ff:fe8e:0/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
osboxes@osboxes:~$
```

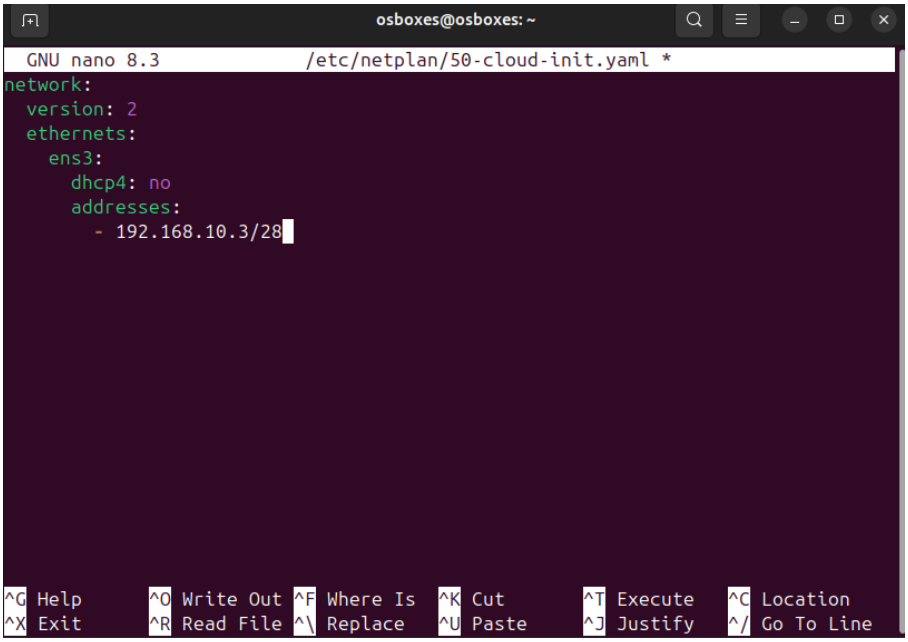
Admin VM Static IP setup:

The 50-cloud-init.yaml file is opened for editor to apply netplan [12].

```
osboxes@osboxes:~$ sudo nano /etc/netplan/50-cloud-init.yaml
```

The file is written as:

Here, the interface is also ens3 here.



Apply:

After saving, the netplan has been applied.

```
osboxes@osboxes:~$ sudo netplan apply

** (generate:6290): WARNING **: 18:14:11.832: Permissions for /etc/netplan/01-ne
twork-manager-all.yaml are too open. Netplan configuration should NOT be accessi
ble by others.

** (generate:6290): WARNING **: 18:14:11.834: Permissions for /etc/netplan/50-cl
oud-init.yaml are too open. Netplan configuration should NOT be accessible by ot
hers.

** (process:6288): WARNING **: 18:14:13.671: Permissions for /etc/netplan/01-net
work-manager-all.yaml are too open. Netplan configuration should NOT be accessib
le by others.

** (process:6288): WARNING **: 18:14:13.672: Permissions for /etc/netplan/50-clo
ud-init.yaml are too open. Netplan configuration should NOT be accessible by oth
ers.

** (process:6288): WARNING **: 18:14:15.773: Permissions for /etc/netplan/01-net
work-manager-all.yaml are too open. Netplan configuration should NOT be accessib
le by others.

** (process:6288): WARNING **: 18:14:15.774: Permissions for /etc/netplan/50-clo
ud-init.yaml are too open. Netplan configuration should NOT be accessible by oth
ers.
osboxes@osboxes:~$
```

Verify:

In the Web VM, it can be seen that the 192.168.10.3/28 IP is used.

```
osboxes@osboxes:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP grou
p default qlen 1000
    link/ether 0c:02:8b:75:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    altname enx0c028b750000
    inet 192.168.10.3/28 brd 192.168.10.15 scope global noprefixroute ens3
       valid_lft forever preferred_lft forever
    inet6 fe80::e02:8bff:fe75:0/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
osboxes@osboxes:~$
```

This means that the desired IP is set in the Admin VM.

2.5. Checking the connectivity between them

This step has verified that the switch, IP configuration, and /28 subnet are all working properly, and that two hosts on the same subnet can usually reach each other via ICMP Echo (ping) [7].

Admin to Web:

```
osboxes@osboxes:~$ ping 192.168.10.2
PING 192.168.10.2 (192.168.10.2) 56(84) bytes of data.
64 bytes from 192.168.10.2: icmp_seq=1 ttl=64 time=0.054 ms
64 bytes from 192.168.10.2: icmp_seq=2 ttl=64 time=0.030 ms
64 bytes from 192.168.10.2: icmp_seq=3 ttl=64 time=0.031 ms
64 bytes from 192.168.10.2: icmp_seq=4 ttl=64 time=0.031 ms
64 bytes from 192.168.10.2: icmp_seq=5 ttl=64 time=0.029 ms
64 bytes from 192.168.10.2: icmp_seq=6 ttl=64 time=0.028 ms
^C
--- 192.168.10.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5077ms
rtt min/avg/max/mdev = 0.028/0.033/0.054/0.009 ms
osboxes@osboxes:~$
```

Web to Admin:

```
osboxes@osboxes:~$ ping 192.168.10.3
PING 192.168.10.3 (192.168.10.3) 56(84) bytes of data.
64 bytes from 192.168.10.3: icmp_seq=1 ttl=64 time=1.44 ms
64 bytes from 192.168.10.3: icmp_seq=2 ttl=64 time=2.39 ms
64 bytes from 192.168.10.3: icmp_seq=3 ttl=64 time=2.20 ms
64 bytes from 192.168.10.3: icmp_seq=4 ttl=64 time=0.844 ms
64 bytes from 192.168.10.3: icmp_seq=5 ttl=64 time=2.32 ms
^C
--- 192.168.10.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 0.844/1.838/2.388/0.602 ms
osboxes@osboxes:~$
```

The Perfect Reply message indicates that Switching is OK.

2.6. Making sure the web server is accessible from the Admin VM (using curl)

From Admin VM:

```
osboxes@osboxes:~$ curl http://192.168.10.2
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"
>
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
    <style type="text/css" media="screen">
  * {
    margin: 0px 0px 0px 0px;
    padding: 0px 0px 0px 0px;
  }

  body, html {
    padding: 3px 3px 3px 3px;

    background-color: #D8DBE2;

    font-family: Ubuntu, Verdana, sans-serif;
    font-size: 11pt;
    text-align: center;
  }

  div.main_page {
    position: relative;
    display: table;
```

The default Nginx index page HTML is visible as <html>, indicating that Nginx on the Web VM is working properly and is accessible from the Admin.

From Web VM:

```
osboxes@osboxes:~$ curl http://192.168.10.3
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"
>
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
    <style type="text/css" media="screen">
  * {
    margin: 0px 0px 0px 0px;
    padding: 0px 0px 0px 0px;
  }

  body, html {
    padding: 3px 3px 3px 3px;

    background-color: #D8DBE2;

    font-family: Ubuntu, Verdana, sans-serif;
    font-size: 11pt;
    text-align: center;
  }

  div.main_page {
    position: relative;
    display: table;
```

Show Apps

The default Nginx index page HTML is visible as <html>, indicating that Nginx on the Web VM is working properly and is accessible from the Admin.

Task 3 - Routing

3.1. Select a virtual Routing solution (Gateway) such as Mikrotik

Router Port Mapping:

Router Port	Connection	Duty
ether3	Cloud (virbr0)	WAN / Internet
ether2	Internal Switch & Worker PC	Worker LAN segment
ether1	External switch + (Web & Admin)	Main LAN

- **2 separate LANs (192.168.10.0/28 and 192.168.20.0/28)**
- **1 WAN (virbr0 → 192.168.122.0/24 NAT network) [14], [15]**

IP Setup

LAN 1 (ether2): Web + Admin VM

Device	IP	Mask Gateway
Gateway (ether1)	192.168.10.1 /28	
Web	192.168.10.2 /28	192.168.10.1
Admin	192.168.10.3 /28	192.168.10.1

LAN 2 (ether3): Worker VMPC

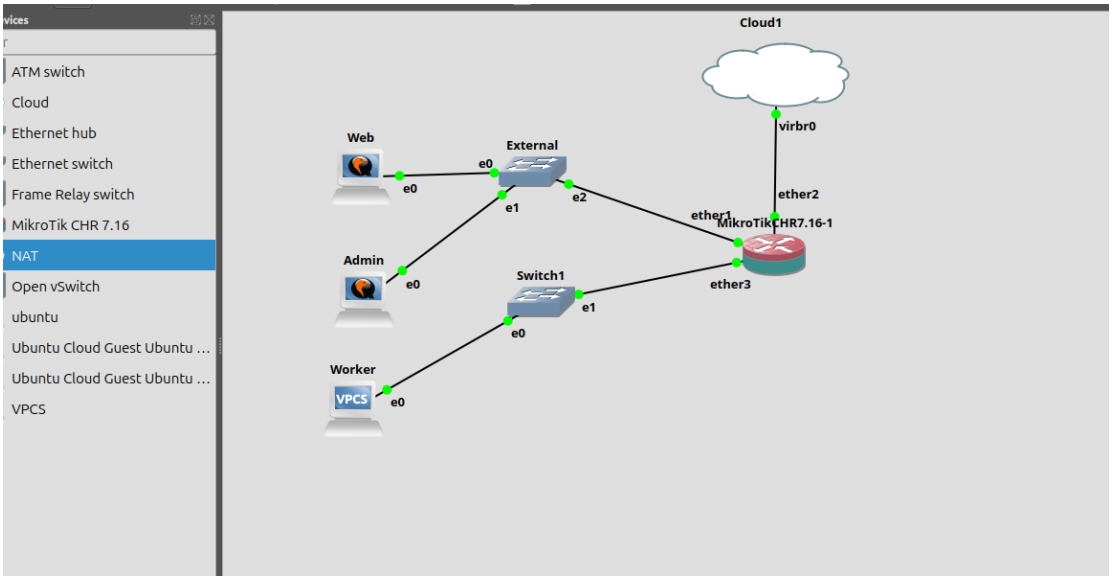
Device	IP	Gateway
Gateway (ether2)	192.168.20.1/28	
Worker PC	192.168.20.2/28	192.168.20.1

WAN (ether2): virbr0

The Ether2 interface gets an IP from DHCP. Typically, in the 192.168.122.x/24 range [15]. These are connected and configured in GNS3.

ether2 : Cloud (virbr0) - WAN

Drag Cloud1 from Devices and select virbr0 and add it. Then connect Cloud virbr0 to Gateway (Router) ether2 with cable. virbr0 is libvirt's NAT network, from here the router's ether3 interface gets the connection to the Internet/NAT side [15]. ether2 does not connect to any WAN and will not get a DHCP IP. Internet cannot be provided through NAT since this Task 3 requirement cannot use NAT.



3.2. Connecting Gateway to the internet and to workstation/host

ether2: External switch (Admin + Web)

In External switch:

- Web to Switch
- Admin to Switch
- Switch to Gateway ether2

This is the Main LAN (192.168.10.0/28) segment.

In ether1 : Worker LAN

- Worker to Internal Switch to ether1

This is the 192.168.20.0/28 segment.

Router (MikroTik) Configuration

The IP has been assigned to the interface using /ip address add in RouterOS on the Gateway (MikroTik) terminal [16], [17].

LAN 1 (ether1):

```
[admin@MikroTik] > /ip address add address=192.168.10.1/28 interface=ether1
```

/ip address add creates a new IP address entry. *address=192.168.10.1/*28 is the IP + subnet mask (CIDR*). interface=ether1*, here, this IP will be set on the ether1 interface [16], [17]. This IP (192.168.10.1) will be the default gateway for the Web + Admin LAN.

LAN 2 (ether2):

```
[admin@MikroTik] > /ip address add address=192.168.20.1/28 interface=ether2
```

The ether2 interface is assigned the gateway IP of the second LAN (192.168.20.1/28) [17]. All hosts in the worker LAN (192.168.20.0/28) use this IP as their gateway.

WAN (ether3):

```
[admin@MikroTik] > /ip dhcp-client add interface=ether3 disabled=no
```

/ip dhcp-client add creates a new DHCP client entry. *interface=ether3*, ether3 connected to virbr0 does not have an IP, gateway, DNS from DHCP. *disabled=no*, DHCP client is enabled immediately [18]. virbr0 network usually gives IP from 192.168.122.0/24, using DHCP [15].

NAT (LAN to Internet):

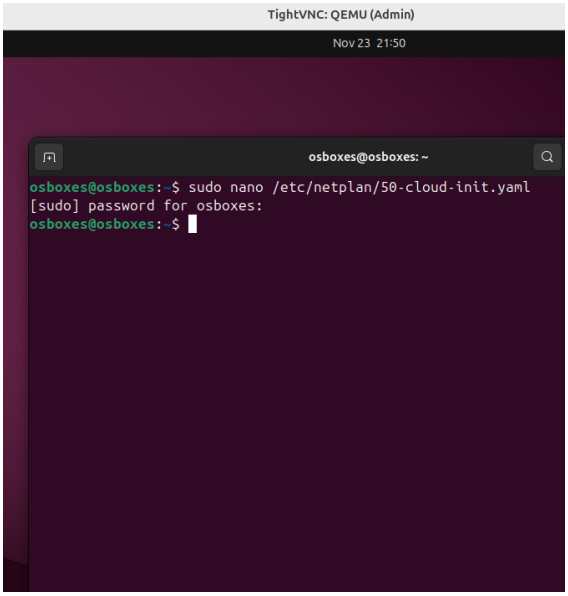
MikroTik uses *srcnat* and *masquerade* to route outgoing traffic from LAN to WAN to the Internet [19], [20]. */ip firewall nat add* adds a new NAT rule and *chain=srcnat* does source NAT on outgoing packets. *out-interface=ether3* is used to go out through ether3. So, this rule applies and *action=masquerade* replaces the source address with the router's WAN IP [19].

```
[admin@MikroTik] > /ip firewall nat add chain=srcnat out-interface=ether3 action=masquerade
```

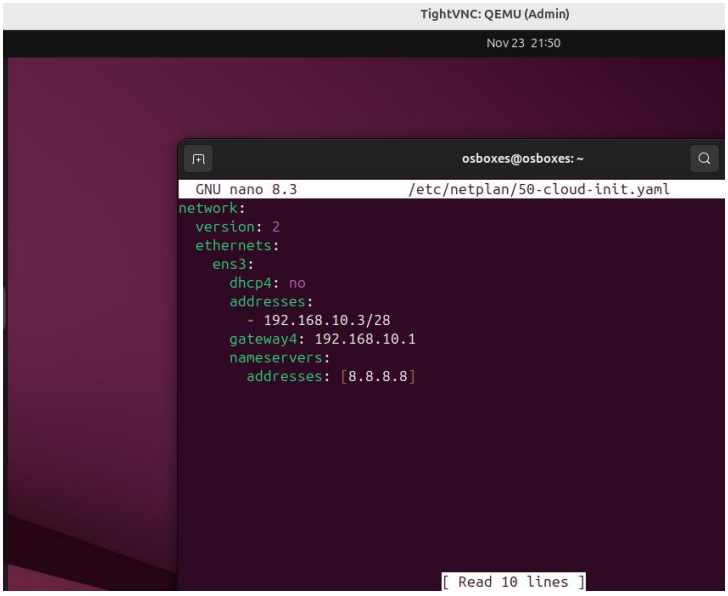
3.4. Setup the gateway for Admin, Web and Worker, then check their connectivity

Web + Admin VM Static IP Setup

Opening the yaml file to apply Admin PC netplan:



Admin VM netplan:



network: is the root node of the netplan config and *version: 2* uses netplan schema v2. *ens3:* is the name of the actual interface where ip a is seen on the machine [21]. *dhcp4:* no means do not use DHCP, IPv4 is static. *addresses: - 192.168.10.3/28*, this IP was used as + subnet mask for the Admin PC. *gateway4: 192.168.10.1* was the router's ether1 IP as default gateway and *nameservers:*

addresses: [8.8.8.8] is Google DNS as DNS [21], [22]. The Admin VM can only talk within the local subnet (192.168.10.0/28), not to other networks/internet.

Apply:

The following command is used to read the config from */etc/netplan/*.yaml* and apply it to NetworkManager.

```
osboxes@osboxes:~$ sudo netplan apply
```

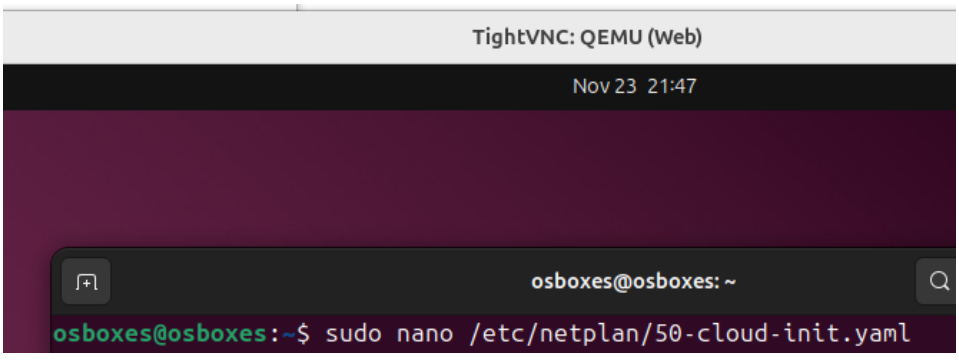
Check Ip:

Here *inet 192.168.10.3/28* is visible on the ens3 interface.

```
osboxes@osboxes:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN gro
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel stat
p default qlen 1000
    link/ether 0c:f9:a3:e3:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    altname enx0cf9a3e30000
    inet 192.168.10.3/28 brd 192.168.10.15 scope global noprefixroute e
       valid_lft forever preferred_lft forever
    inet6 fe80::ef9:a3ff:fee3:0/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
osboxes@osboxes:~$
```

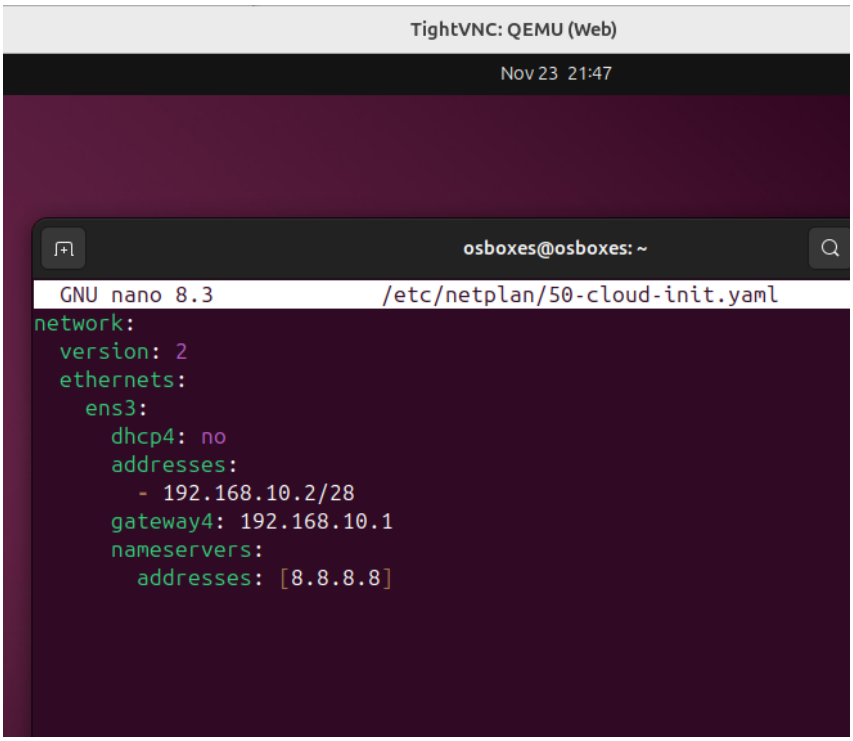
Opening Web PC file:

The yaml file for applying netplan on Web PC is opened using the following command.



Web VM netplan:

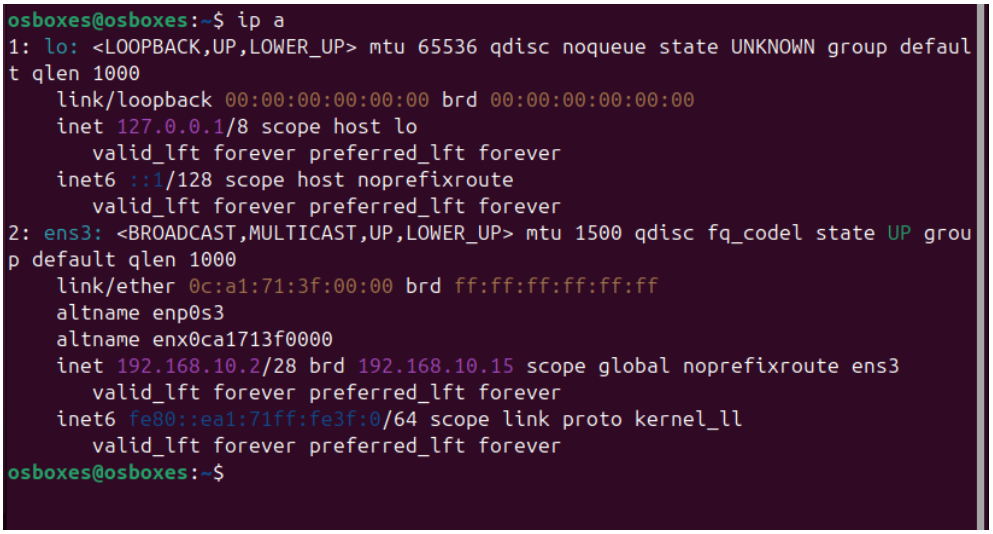
Same configuration as Admin VM but only IP = 192.168.10.2/28 and gateway and DNS are the same here too.



Applying the netplan for Web PC:



Checking Ip:



Worker PC Static IP

Worker (VPCS) console:

IP address = 192.168.20.2 Subnet mask = 255.255.255.240 (means /28)

Default gateway = 192.168.20.1 (Router ether2 IP) [23].

```
PC1> ip 192.168.20.2 255.255.255.240 192.168.20.1
Checking for duplicate address...
PC1 : 192.168.20.2 255.255.255.240 gateway 192.168.20.1
```

Testing Period:

```
PC1> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=64 time=1.707 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=64 time=1.758 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=64 time=1.473 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=64 time=1.534 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=64 time=1.745 ms
```

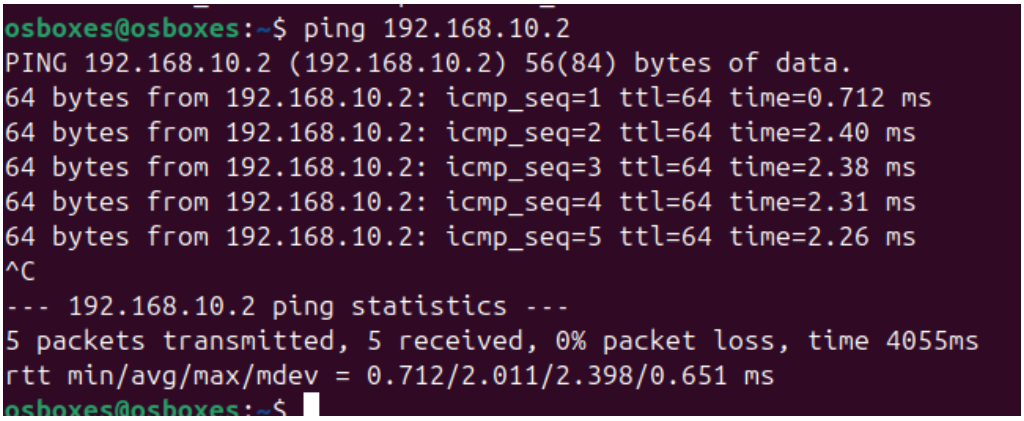
The reply comes in correctly, indicating that the Worker to Router LAN2 connection is working properly.

Inter-LAN Routing Test

Web to Admin:

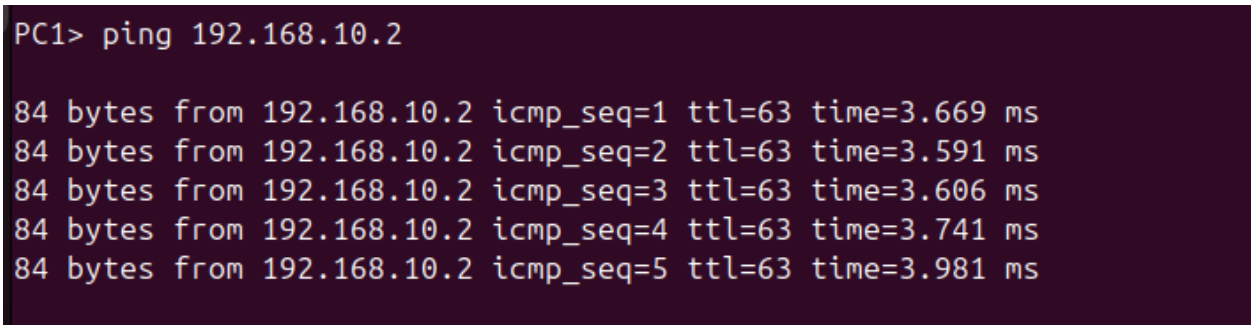
```
osboxes@osboxes: ~
osboxes@osboxes:~$ ping 192.168.10.3
PING 192.168.10.3 (192.168.10.3) 56(84) bytes of data.
64 bytes from 192.168.10.3: icmp_seq=1 ttl=64 time=1.74 ms
64 bytes from 192.168.10.3: icmp_seq=2 ttl=64 time=0.731 ms
64 bytes from 192.168.10.3: icmp_seq=3 ttl=64 time=2.40 ms
64 bytes from 192.168.10.3: icmp_seq=4 ttl=64 time=2.27 ms
64 bytes from 192.168.10.3: icmp_seq=5 ttl=64 time=2.14 ms
^C
--- 192.168.10.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4010ms
rtt min/avg/max/mdev = 0.731/1.855/2.396/0.603 ms
osboxes@osboxes:~$
```

Admin to Web:

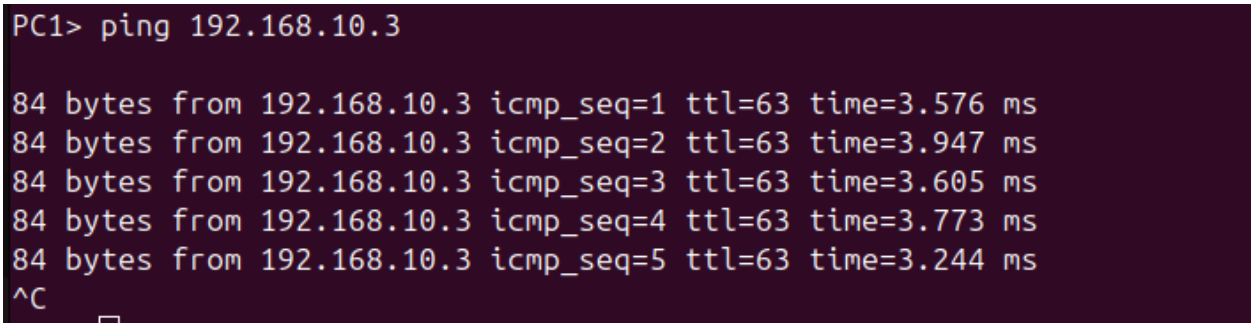


Ping replies are coming, so these are working. This means that /28 IP + netplan static config was ok and Switch + ether1 IP setup was ok.

Worker to Web:



Worker to Admin:



Worker *192.168.10…* subnet is not known locally, so the gateway was sending the packet to 192.168.20.1. In the router's routing table, both are connected networks and forwarded from one LAN to the other [14]. Everything is working because the router is routing two LANs.

3.5. Configure port forwarding for http and ssh to Web and Admin respectively

When entering from virbr0 direction:

- TCP port 80 - Web VM (192.168.10.2:80)
- TCP port 22 - Admin VM (192.168.10.3:22)

In RouterOS, dstnat is set like this [19].

HTTP to Web (80 for 192.168.10.2)

```
[admin@MikroTik] > /ip firewall nat add chain=dstnat dst-port=80 protocol=tcp
\...    action=dst-nat to-addresses=192.168.10.2 to-ports=80
```

chain=dstnat, works on incoming (destination NAT) traffic and *dst-port=80 protocol=tcp* are all requests coming in on TCP port 80. *action=dst-nat* is used to change the destination IP/Port and sends it to the HTTP port of the Web VM via *to-addresses=192.168.10.2 to-ports=80*. When I accessed the router's IP:80 from virbr0, it went to the Web VM's web server.

SSH to Admin (22 for 192.168.10.3)

```
[admin@MikroTik] > /ip firewall nat add chain=dstnat dst-port=22 protocol=tcp
\...    action=dst-nat to-addresses=192.168.10.3 to-ports=22
```

Port 22 is TCP traffic coming in over SSH and forwarded from the router's WAN IP to the Admin VM (*192.168.10.3:22*). This setup allows SSH into the Admin VM from the WAN side, while everything works on the private LAN [19].

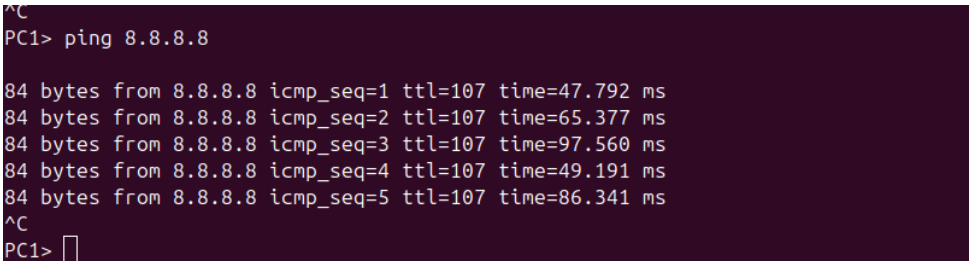
3.6. Check that you can ssh to the Admin and access your web page from your workstation/host.

Admin VM to google.com:

```
osboxes@osboxes:~$ ping google.com
PING forcesafesearch.google.com (216.239.38.120) 56(84) bytes of data.
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=1 ttl=106 time=41.1 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=2 ttl=106 time=43.9 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=3 ttl=106 time=43.1 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=4 ttl=106 time=43.5 ms
^C
--- forcesafesearch.google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 41.144/42.917/43.904/1.064 ms
```

The reply comes perfectly and it means that from Admin VM to gateway to NAT to Internet, all are ok and DNS-8.8.8.8 is fine.

Worker PC:



Reply is received when pinging and packets are being sent correctly from Worker LAN to router to NAT [23].

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